

# Geopolitical Impunity and Extractive Activity

A Hybrid Physics-ML Framework for Detecting  
Mining Anomalies in Disputed Territories

PhD Research Proposal | Data Science & Environmental Geopolitics

**Core Hypothesis:** Contested sovereignty creates enforcement vacuums. The resulting illegal extraction can be quantified in the residual between physical hydrological models and satellite observations.



## The Political Mechanism

Contested Sovereignty

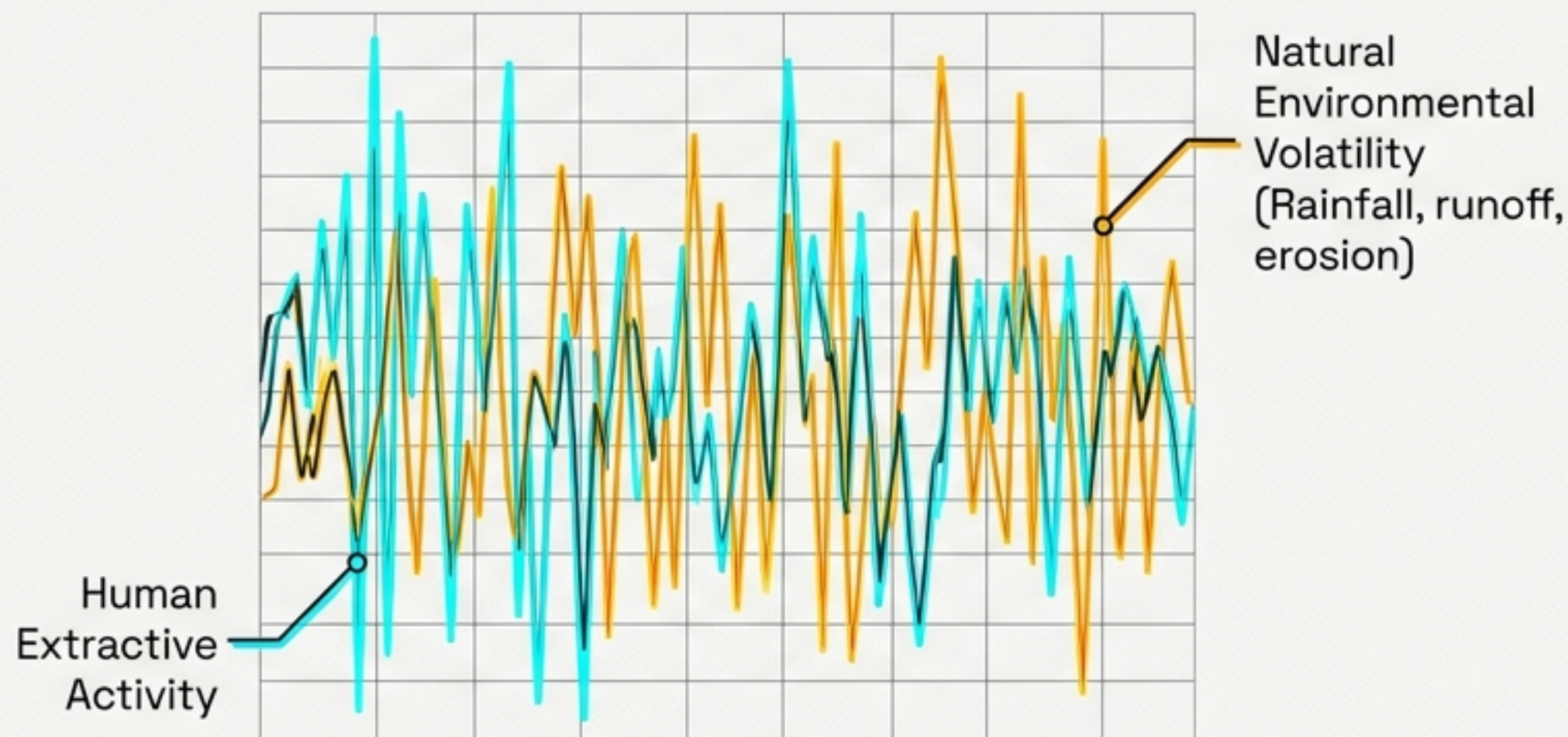


Enforcement Vacuum



Unregulated Extraction  
(Mining/Deforestation)

## The Measurement Problem



**Current limitation:** Standard remote sensing and aggregate statistics fail to isolate the human extraction signal from natural hydrological noise in contested areas.



### RQ0 [The Baseline]

What is the expected natural water turbidity and sediment load based purely on hydrology and topography?

### RQ1 [The Anomaly]

Do contested territories exhibit systematic excess extractive activity beyond natural predictions?

### RQ2 [The Drivers]

Is the magnitude of this excess statistically predicted by geopolitical covariates (governance, prices, borders)?

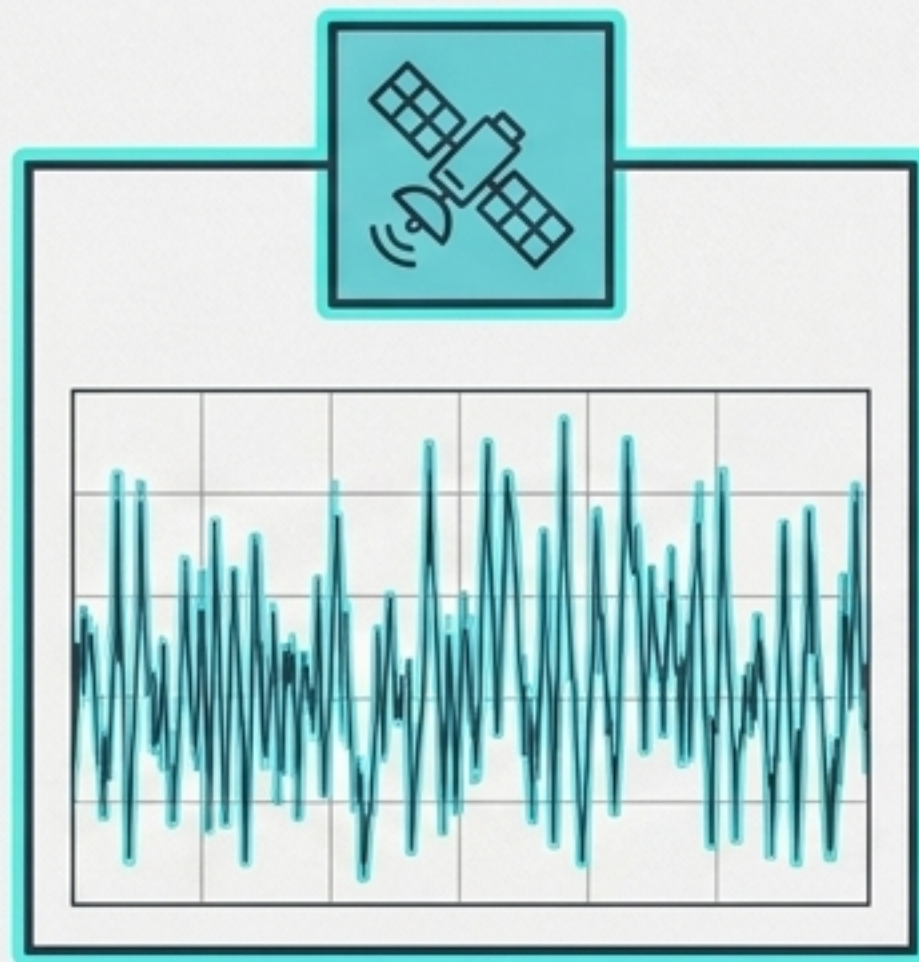
### RQ3 [The Mechanism]

Do extraction anomalies spike and persist following specific dispute escalation events?



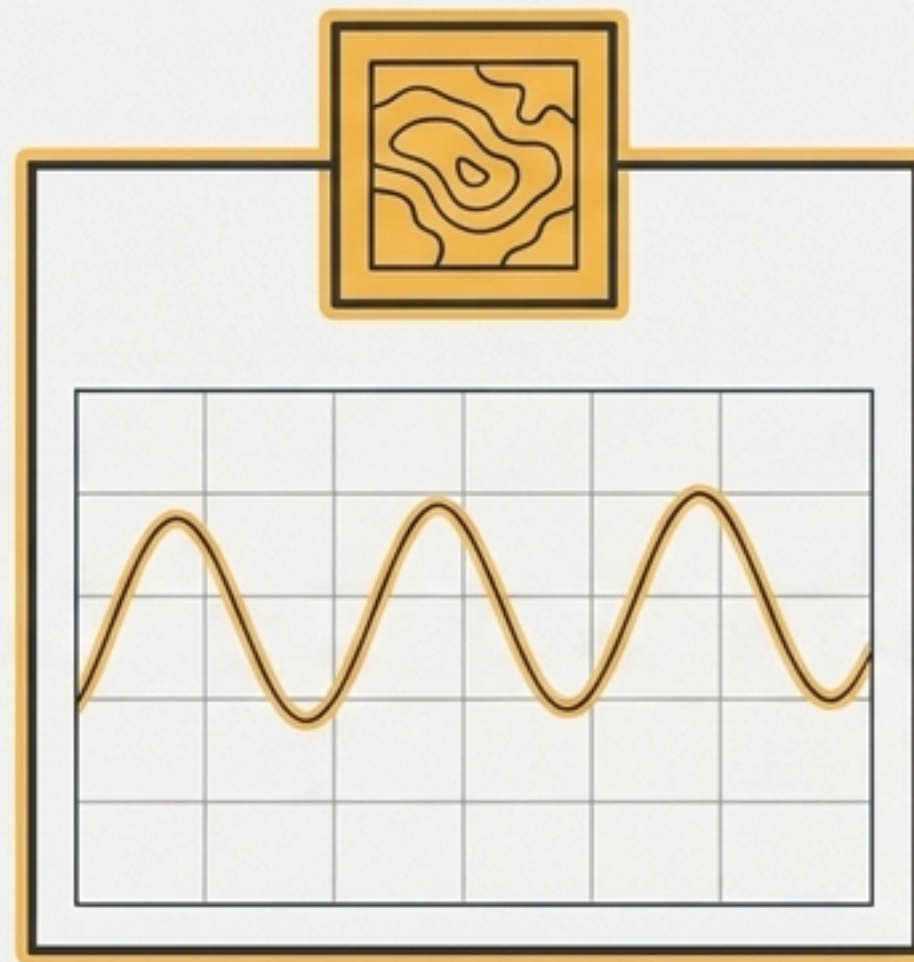
|                      | Pure Remote Sensing                      | Pure Political Economy                         | Proposed Physics-ML Hybrid                             |
|----------------------|------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Method               | Pure ML / Change Detection               | Aggregate Country-Level Stats                  | Catchment-Level Time Series                            |
| Baseline             | Historic Satellite Averages              | Economic Averages                              | Interpretable<br>Physical Baseline                     |
| Geopolitical Framing | Often Absent or Secondary                | Primary Focus                                  | Framing the Residual as the Geopolitical Signal        |
| Drawbacks / Value    | Confounded by natural climate variations | Too coarse for border-zone localized conflicts | Requires high data synthesis and cross-domain pipeline |





y\_observed  
(Sentinel-2 NDTI  
Turbidity)

—



y\_physics  
(SWAT Baseline Prediction)

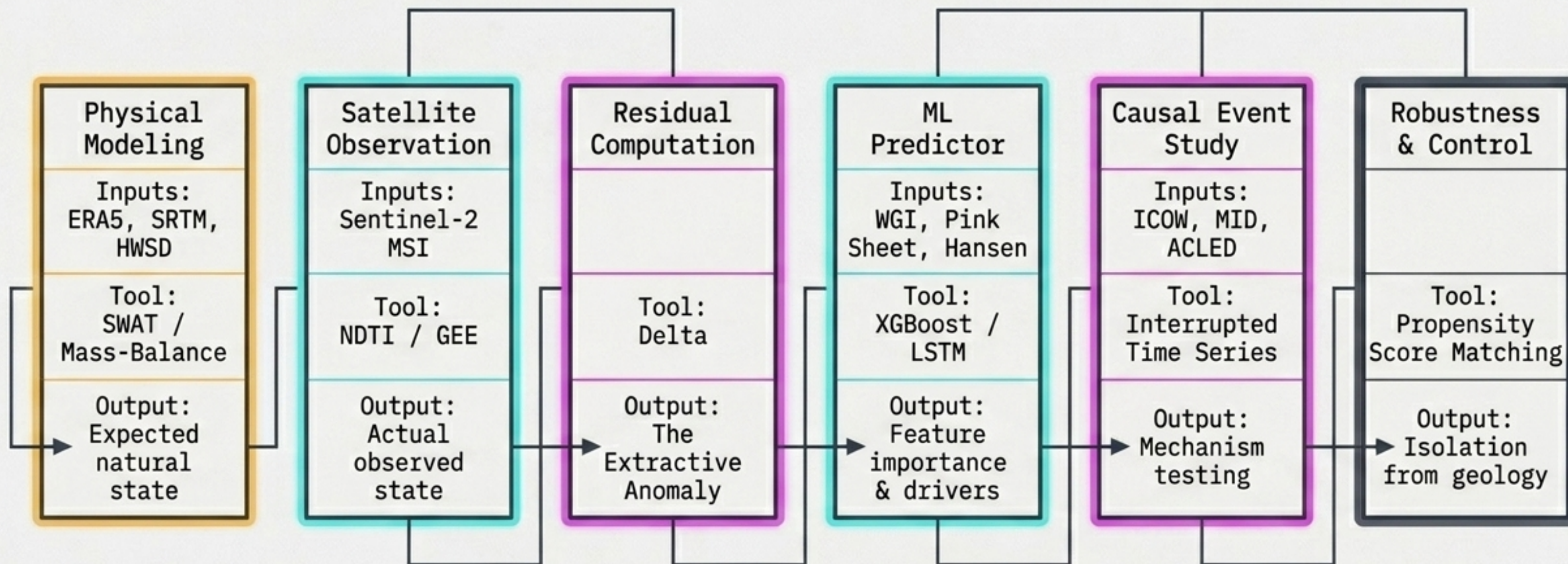
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The Residual

Positive residuals equal excess sediment. This is the anomaly signal – the fingerprint of human activity that physics cannot account for.







## Hydrology Data

|                             |           |
|-----------------------------|-----------|
| ERA5 (Precipitation)        | Free/Open |
| SRTM 30m (DEM/Slope)        | Free/Open |
| FAO HWSD (Soil Erodibility) | Free/Open |

## Remote Sensing Data

|                               |           |
|-------------------------------|-----------|
| Sentinel-2 (Turbidity/NDTI)   | Free/Open |
| ESA CCI (Baseline Land Cover) | Free/Open |
| Hansen/GFW (Tree Cover Loss)  | Free/Open |

## Geopolitics Data

|                                  |           |
|----------------------------------|-----------|
| ICOW (Sovereignty Disputes)      | Free/Open |
| MID Dataset (Militarized Events) | Free/Open |
| WGI (Governance/Rule of Law)     | Free/Open |

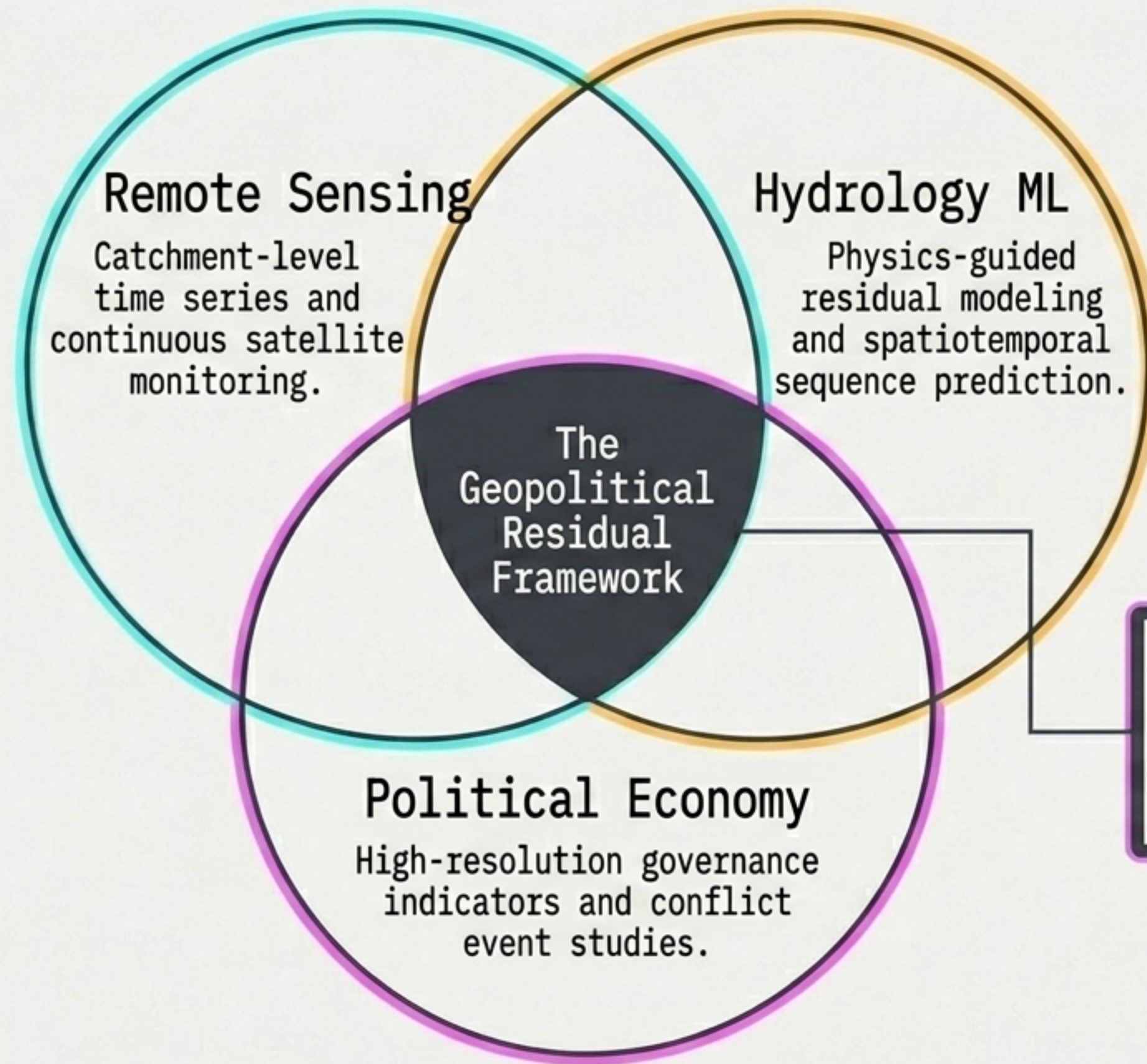
## Economic Data

|                                      |           |
|--------------------------------------|-----------|
| World Bank Pink Sheet (Metal Prices) | Free/Open |
| VIIRS (Nighttime Lights)             | Free/Open |

## Tech Stack

Python (rasterio, shap) — R (spdep) — Google Earth Engine — SWAT/QGIS

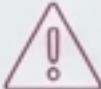
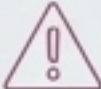
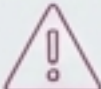




Framing hydrological residuals as a quantifiable measure of geopolitical impunity.



# Methodological Diagnostics: Risk vs. Mitigation

| Known Methodological Risk                                                           |                                                                                                             | Proposed Mitigation                                                                   |                                                                                                               |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|    | SWAT calibration requires stream gauge data, which is sparse in contested zones.                            |    | Utilize uncalibrated/regionalized parameters; review literature on ungauged basin estimation.                 |
|    | Satellite turbidity indices only detect surface/alluvial mining, missing underground hard-rock mines.       |    | Strictly and explicitly bound the research scope to surface/alluvial extraction impacts.                      |
|  | Dispute event databases (ICOW, MID) have variable temporal granularity.                                     |  | Pre-screen and validate temporal coverage before committing to the causal event study design.                 |
|  | Geological Endogeneity (disputed zones may naturally have more minerals, driving both conflict and mining). |  | Apply Propensity Score Matching comparing disputed vs. non-disputed zones with identical geological profiles. |



Phase 1: Next 30 Days



**Validate Satellite Data Pipeline.**  
Execute Google Earth Engine  
Sentinel-2 NDTI scripts to  
confirm baseline turbidity  
detection at 10m resolution.

Phase 2: Next 60 Days



**Build Single-Basin Toy Model.**  
Test the SWAT/mass-balance  
baseline on a single, well-  
documented contested catchment to  
isolate an initial residual.

Phase 3: Next 90 Days



**Literature Deep-Dive.**  
Synthesize methods on  
ungauged basin estimation and  
spatial panel econometrics.

**Discussion & Architecture Review**